

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/264389958>

High force development augments skeletal muscle signalling in resistance exercise modes equalized for time under tension

Article in *Pflügers Archiv - European Journal of Physiology* · July 2014

DOI: 10.1007/s00424-014-1579-y

CITATIONS

12

READS

265

10 authors, including:



Sebastian Gehlert

Deutsche Sporthochschule Köln

40 PUBLICATIONS 137 CITATIONS

[SEE PROFILE](#)



Daniel Jacko

Deutsche Sporthochschule Köln

7 PUBLICATIONS 15 CITATIONS

[SEE PROFILE](#)



Axel Johannes Knicker

Deutsche Sporthochschule Köln

49 PUBLICATIONS 412 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Mechanoprotection and mechanical stress on human skeletal muscle [View project](#)



Posttranslational Modifications of Skeletal Muscle Ryanodine Receptor 1 by Resistance Exercise [View project](#)

High force development augments skeletal muscle signalling in resistance exercise modes equalized for time under tension

Sebastian Gehlert · Frank Suhr · Katrin Gutsche · Lena Willkomm · Julia Kern · Daniel Jacko · Axel Knicker · Thorsten Schiffer · Henning Wackerhage · Wilhelm Bloch

Received: 18 February 2014 / Revised: 3 July 2014 / Accepted: 8 July 2014
© Springer-Verlag Berlin Heidelberg 2014

Abstract How force development and time under tension (TUT) during resistance exercise (RE) influence anabolic signalling of skeletal muscle is incompletely understood. We hypothesized that high force development during RE is more important for post-exercise-induced signalling than submaximal and fatiguing RE with lower force development but similar TUT. Twenty-two male subjects (24 ± 6 years, 181 ± 9 cm, 79 ± 2 kg) performed three distinct RE modes in the fed state with equal TUT but distinct force output: (i) maximal eccentric RE (ECC, $n=7$) three sets, eight reps, 100 % eccentric dynamic force; (ii) standard RE (STD, $n=7$), three sets, 10 reps, 75 % dynamic force; and (iii) high fatiguing single-set RE (HIT, $n=8$), 20 reps, 100 % eccentric-concentric force; vastus lateralis biopsies were collected at baseline, 15, 30, 60, 240 min and 24 h after RE,

and the signalling of mechanosensitive and mammalian target of rapamycin (mTOR)-related proteins was determined. The phosphorylation levels of pFAK^{Tyr397}, pJNK^{Thr183/Tyr185}, pAKT^{Thr308/Ser473}, pmTOR^{Ser2448}, p4E-BP1^{Thr37/46}, p70s6k^{Thr389/Ser421/Thr424} and pS6^{Ser235/236} were significantly higher in ECC than those in STD and HIT at several time points ($P<0.01$). pJNK^{Thr183/Tyr185} and pS6^{Ser235/236} levels were significantly higher in type II myofibres in ECC compared with STD and HIT. HIT exerted throughout the weakest signalling response. We conclude that high force development during acute RE is superior for anabolic skeletal muscle signalling than fatiguing RE with lower force output but similar TUT. Our results suggest that this response is substantially driven by the higher activation of type II myofibres during RE.

Electronic supplementary material The online version of this article (doi:10.1007/s00424-014-1579-y) contains supplementary material, which is available to authorized users.

S. Gehlert (✉) · F. Suhr · K. Gutsche · L. Willkomm · J. Kern · D. Jacko · W. Bloch
Institute of Cardiology and Sports Medicine, Department of Molecular and Cellular Sport Medicine, German Sport University Cologne, Am Sportpark Müngersdorf 6, 50933 Cologne, Germany
e-mail: gehlert@dshs-koeln.de

F. Suhr
e-mail: suhr@dshs-koeln.de

K. Gutsche
e-mail: katring@web.de

L. Willkomm
e-mail: l.willkomm@dshs-koeln.de

J. Kern
e-mail: juliakern@web.de

D. Jacko
e-mail: d.jacko@dshs-koeln.de

W. Bloch
e-mail: w.bloch@dshs-koeln.de

A. Knicker
Institute of Movement and Neurosciences, German Sport University Cologne, Cologne, Germany
e-mail: knicker@dshs-koeln.de

T. Schiffer
Outpatient Clinic for Sports Traumatology, German Sport University Cologne, Cologne, Germany
e-mail: t.schiffer@dshs-koeln.de

H. Wackerhage
School of Medical Sciences, Foresterhill, University of Aberdeen, Aberdeen, Scotland, UK
e-mail: h.wackerhage@abdn.ac.uk

Keywords Resistance exercise · Skeletal muscle signalling · mTOR signalling · Myofibre types · Mechanical stress · Protein synthesis

Abbreviations

Nm	Newton meter
RE	Resistance exercise
REPS	Repetitions
TUT	Time under tension
AKT	v-AKT murine thymoma viral oncogene
4E-BP1	Eukaryotic initiation factor 4E-binding protein 1
MAPK	Mitogen-activated protein kinase
mTOR	Mammalian target of rapamycin
JNK	c-jun N-terminal kinase
p70S6K	70 kDa S6 protein kinase
p90RSK	90 kDa ribosomal S6 protein kinase
S6	S6 ribosomal protein
FAK	Focal adhesion kinase
AMPK	5-AMP-activated protein kinase
GSK3- β	Glycogen synthase kinase 3 beta
TSC-2	Tuberous sclerosis complex 2 (Tuberin)
CK	Creatine kinase

Introduction

Resistance exercise (RE) increases anabolic skeletal muscle signalling and protein synthesis acutely and for up to several days after exercise [1]. Protein synthesis responses after RE critically depend on the mammalian target of rapamycin (mTOR) pathway, as its inhibition via rapamycin prevents the exercise-induced increase of protein synthesis in human skeletal muscle [2]. The phosphorylation of several of the activity-related proteins of the mTOR pathway, specifically TSC-2 [3], mTOR [4], p70s6k [5] and 4E-BP1 [6], is observed in the acute phase following RE, and the magnitude of initial mTOR-related signalling after RE correlates with increased protein synthesis and skeletal muscle hypertrophy [5, 7]. Consequently, the phosphorylation pattern of activity-related sites of the mTOR network is frequently investigated to assess the efficiency of various RE types and RE determinants in inducing early states of skeletal muscle adaptation [8, 9]. However, although critical regulators of muscle growth in RE are widely determined, the most effective combination of force development (also described as exercise intensity), training volume and time under tension (TUT) for the stimulation of skeletal muscle growth is still under debate [10]. Both high force development in RE [11] and high training volumes [12] induce higher p70s6k and S6 phosphorylation levels compared with submaximal force generation and lower training volumes, respectively. Additionally, it has been shown that an elevated time course in which a low force of

skeletal muscle is generated (43 s with 30 % of 1 repetition maximum) stimulates the phosphorylation of mTOR, p70s6k and 4E-BP1 more than RE with high force development (16 s with 90 % of 1 RM) but lower tension time [13]. However, in this study, increased TUT but low force was created by applying a five times higher count of repetitions and consequently training load compared with high force but low TUT (24 vs. 5 repetitions). Hence, the influence of force generation on skeletal muscle signalling responses under equal loading time could not be determined by using their protocol. In contrast to the influence of high TUT on skeletal muscle signalling, it has also been demonstrated that maximal eccentric contractions, which are associated with a higher mechanical force output than concentric contractions, induce higher phosphorylation levels of AKT-mTOR and mitogen-activated protein kinase (MAPK) signalling than submaximal eccentric and maximal concentric contractions [11]. This finding was determined especially in type II fibres [9], and it further underlines force development of skeletal muscle during RE to be of vital importance for directed and fibre-type specific skeletal muscle adaptation. Since skeletal muscle is sensitive to both TUT and the magnitude of skeletal muscle tension, it can be hypothesized that high force development induced by high RE intensities exerts higher signalling responses compared with less intensive RE modes when similar time courses of TUT are applied. Indeed, it has already been determined in rat skeletal muscle that MAPK signalling is highly sensitive to the time tension integral [14] of mechanical loading. Despite the aforementioned findings which attribute TUT as well as magnitude of force generation major roles in determining acute skeletal muscle responses, we assume that the interaction between force generation and TUT during RE lacks a sufficient description. To test whether high force development of skeletal muscle under equalized TUT is a key variable for the induction of acute signalling events in skeletal muscle, we conducted a study in which 22 male subjects were randomly assigned to three divergent modes of one-legged isokinetic knee extensions (i.e. maximal eccentric, ECC; submaximal concentric-eccentric, STD; high intense single set to fatigue, HIT) with distinct exercise intensity and force generation but equalized TUT. As an estimate of the resultant muscle growth-stimulating effect, we determined the phosphorylation of activity-related sites of the mTOR signalling pathway as well as focal adhesion kinase (FAK) and c-jun N-terminal kinase (JNK), which have been identified as reliable readouts for the magnitude of mechanical stress [15, 16].

We explicitly aimed to investigate the acute cellular impact of distinct RE modes via essential signalling measures that allowed us to assess the acute efficiency of RE determinants in inducing early adaptational responses of skeletal muscle. We hypothesized that higher force generation and mechanical demands of skeletal muscle induce higher signalling responses than RE modes with lower force generation when

carried out over equalized TUT. Specifically, we hypothesized that subjects who perform ECC show the highest force development and phosphorylation levels of signalling proteins followed by STD and HIT subjects, which together represent a lower force output. We also assumed skeletal muscle signalling responses to be present in single type I and II myofibres and that type II fibre signalling is the highest in ECC, as eccentric exercise particularly leads to the increased recruitment of this fibre type [17]. Extended knowledge concerning effective RE regimens is important for maximizing training effects under conditions of limited training time and to emphasize growth-stimulating effects in defined myofibre proportions such as type II fibres. In particular, this fibre type is highly susceptible to atrophy under conditions of immobilization, space flight and in the elderly.

Methods

Subjects

Twenty-two healthy male sport students (age 24 ± 6 years, height 181 ± 9 cm and weight 79 ± 2 kg) were recruited and participated in the present study. All subjects were non-smokers and not taking any medication. Prior study participation, subjects were screened for health with clinical history, physical examination and laboratory tests, including blood count and ECG. We explicitly aimed to investigate trained subjects and chose male sport students that regularly attended physical activity courses at the university and were experienced with RE (RE participation up to two times per week). Participants were informed orally and in writing of the study's purpose and the possible risks involved before providing written informed consent to participate. The study was approved by the Ethic Committee of the German Sport University Cologne in compliance with the Declaration of Helsinki. Subjects were assigned to an anonymous ID during the study and randomly assigned to one of three different exercise regimens: HIT ($n=8$), STD ($n=7$) and ECC ($n=7$).

Standardization of diet and activity before exercise

Subjects were instructed to refrain from RE 14 days prior to this study and from any physical activity 48 h prior to baseline biopsies, exercise testing and the main RE stimulus. The day prior to the resting biopsy as well as the main exercise stimulus, a standardized protein energy drink (Fresubin® protein energy drink, Fresenius Kabi Deutschland GmbH, Bad Homburg, Germany; containing 20 g protein, 24.8 g carbohydrate, 13.4 g fat, providing 1,260 kJ) was provided to the subjects at 22:00 which afterwards fasted overnight. The following morning, 1 h prior to the resting biopsy and the main RE intervention, subjects were advised to drink a second energy

drink in order to carry out the exercise in the fed state. Subjects were allowed to drink water ad libitum during all exercise interventions.

Exercise testing

Unilateral isokinetic leg extension with visual feedback was chosen as RE model (ISOMED 2000 D&R Ferstl GmbH, Hemau, Germany) allowing to precisely standardize range of motion (ROM) and movement speed of each contraction. At two different time points, 9 and 7 days prior to the main exercise stimulus, maximum dynamic eccentric and concentric force output during single-leg extensions was determined. Prior strength testing, subjects performed a 10-min unspecific warm-up on a Lohde Excalibur (Lohde, Groningen, Netherlands) cycling ergometer with 1 Watt (W) per kilogram bodyweight. Afterwards, subjects were seated on the isokinetic machine, and ROM, knee joint and lever arm axis were adjusted to each subject. Subsequently, participants performed a specific submaximal RE warm-up phase consisting of three sets of four concentric and eccentric repetitions with 90-s rest and submaximal intensity. After 3-min resting, subjects of the HIT and STD group were subjected to a single set of ten repetitions with concentric and eccentric knee extensions. After the first seven submaximal repetitions, subjects were advised to carry out three maximal repetitions during the whole ROM in the concentric and eccentric phase. All maximum strength tests were carried out with equal movement speed as applied for the specific exercise mode in the main exercise stimulus. Subjects assigned to ECC carried out the equal strength test as subjects in STD and HIT with solely eccentric contractions. In ECC, the exercising leg was returned passively after each eccentric contraction to the starting position for the next repetition hereby avoiding concentric contractions. Due to the force-length and force-angle relationship, force development during concentric and eccentric contractions followed a hyperbolic curve. The highest maximum dynamic force curve during strength tests was applied for the main RE mode as the 100 % reference curve in ECC and HIT and to 75 % for subjects of STD.

Experimental trials

On the day of the main trial, subjects reported to the laboratory at 07:45 AM. Prior to the RE bouts, a 10-min warm-up with a workload of 1 W/kg on a cycle ergometer was performed (see "Exercise testing"). Following that, a combined warm-up and main exercise program were carried out on the isokinetic leg extension machine. The specific warm-up phase featured three sets of four submaximal (30 % of maximum force) concentric and eccentric repetitions and 90-s resting intervals between

the sets. The angular velocity was determined by 65°/s movement speed in concentric and eccentric phases for each subject and exercise group. After a 3-min resting phase, the specific main RE mode was applied.

Resistance exercise stimuli

All exercise regimens were carried out as unilateral single-leg extensions with 70° ROM.

- **STD:** Three sets of ten unilateral concentric and eccentric leg extensions with 3-min rest between sets and 75 % of maximum eccentric and concentric force. Movement speed 65°/s with 66 s TUT.
- **HIT:** One single set of twenty unilateral concentric and eccentric leg extensions with the highest possible force generation in every contraction. Movement speed 40°/s with 70 s TUT.
- **ECC:** Three sets of eight unilateral maximum eccentric leg extensions with 3-min rest between sets and 100 % maximum eccentric force in each repetition. Movement speed 25°/s with 67.2 s TUT.

All subjects of the HIT and ECC group were verbally encouraged during the exercise session to perform all contractions in RE modes with maximum voluntary force. A visual feedback of their 100 % force reference curve was shown on a screen for further encouragement during each contraction. For subjects of the SD group, concentric and eccentric power was set to 75 % of their determined 100 % curve.

Blood samples

Venous blood samples were drawn by antecubital venipuncture and collected into heparinized evacuated containers at baseline (PRE), 15, 30, 60, 240 min as well as 24 h post-RE. Containers were stored for 18 min at 4 °C without agitation and then centrifuged at 4,000g for 12 min at 4 °C. The serum supernatant was stored at -20 °C and the pellet discarded. Serum creatine kinase levels (U/L) were determined via photometric determination at 340 nm on a KOBAS Mira Plus chemistry analyzer (Roche GmbH, Mannheim, Germany) using *N*-acetylcystein stabilized creatine kinase test kits (D98485) (DIALAB GmbH, Wiener Neudorf, Austria). All samples of every subject were measured twice on two consecutive measurements in 1 day.

Muscle biopsies

Baseline biopsies of the vastus lateralis muscle were taken at rest 5 days prior to the exercise intervention. On the day of the experimental trial, biopsies were collected 15, 30, 60, 240 min, and 24 h post-exercise from the exercised leg. Biopsies

were taken 2–4 cm apart from each other starting 4 cm cranial of the resting biopsy at 15-min post-exercise. Biopsies of 30 min were taken 2 cm distal from 15-min incision. Biopsy of 60 min was collected at the level of the resting biopsy. Biopsies of 240 min and 24 h were taken 2 cm distal from the resting biopsy each 2 cm distal apart from each other, so that all six biopsies were collected within a range of 8 cm. Muscle biopsy samples (100–150-mg muscle tissue) were obtained 2.5 cm below the fascia using the percutaneous needle biopsy method (needle diameter 5 mm) and collected from the middle region of the vastus lateralis muscle between spina iliaca anterior superior and the lateral part of the patella. Skeletal muscle samples were freed from blood and dissected from any visible fat and connective tissue. Muscle tissue samples for Western blotting analysis were immediately placed in cryotubes and frozen in liquid nitrogen and stored at -80 °C prior to analysis. Muscle samples for immunohistochemical analysis were carefully aligned for cross-sectional analysis, embedded in Tissue-Tek (Sakura Finetek, Zoeterwoude, Netherlands), frozen in liquid nitrogen-cooled isopentane and stored at -80 °C.

Immunohistochemical staining and analysis

Consecutive 7-μm cross sections of each biopsy time point were cut with a LEICA CM7300 Cryostat and mounted in duplicate on Polysine® slides (VWR International GmbH, Darmstadt, Germany) air-dried and stored at -80 °C until further analysis. Cross sections of each biopsy time point of a single subject were mounted on two slides. PRE, 15- and 30-min cross sections as well as 60-min, 240-min and 24-h cross sections were placed on separate slides. All slides which were compared were stained at the same time. Each consecutive cross section was simultaneously stained for myosin heavy chain type I and also for the relevant target protein to determine sarcoplasmic density of pJNK^{Thr183/Tyr185} and pS6^{Ser235/236} separately in types I and II fibres.

The following antibodies and dilutions were used for skeletal muscle immunohistochemistry. Antibodies raised against pS6^{Ser235/236} (1:100) and pJNK^{Thr183/Tyr185} (1:200) were purchased from Cell Signaling (Beverly, MA, USA). A4.951 antibodies (1:200) which are raised against human adult slow myosin heavy chain type I were used from the Developmental Studies Hybridoma Database (Iowa City, USA). Immunohistochemical analysis was conducted as described previously [18]. Briefly, slides containing cross sections were brought to room temperature and subsequently incubated for 10 min at 37 °C and afterwards fixed for 8 min in -20 °C acetone. Sections were blocked in 0.05 mM TBS containing 5 % bovine serum albumin (BSA) for 1 h at room temperature. Slides were then incubated overnight at 4 °C with primary antibodies diluted in 0.8 % BSA. On the following morning slides were rinsed five times for 5 min each with TBS and then incubated for 1 h with appropriate polyclonal biotinylated secondary antibodies (Dako Cytomation, Glostrup,

Denmark) diluted 1:400 in 0.05 mM TBS. Negative controls were processed in the same way as the experimental cross sections; however, primary antibody incubations were substituted by incubation with 0.8 % BSA-TBS. Slides were incubated for 1 h with streptavidin-biotinylated horseradish peroxidase complex (Amersham Biosciences, Uppsala, Sweden) diluted 1:400 in TBS. Staining was carried out using a 3,3'-diaminobenzidine (DAB) solution (0.09-M phosphate buffer (pH 7.4), 2.2 mM DAB, 7.03 mM ammonium chloride, 0.93 mM nickel sulphate, 10.44 mM β -D-glucose and 0.000024 mM glucose oxidase). After dehydration, the stained sections were embedded in Entellan (Merck, Darmstadt, Germany) and supplied with a coverslip. Muscle cross sections from all biopsy time points of subjects and groups were stained within a single batch using the same antibody dilution and development time to minimize variability in staining efficiency between groups and subjects. Sarcoplasmic levels of pS6^{Ser235/236} and pJNK^{Thr183/Tyr185} were determined separately for type I and type II myofibres. Target specific staining intensity for each myofibre was quantified by selection of the sarcoplasmic region of the myofibre and its subsequent measurement by optical densitometry using the software ImageJ® (National Institutes of Health, USA).

Western blotting

Twenty milligrams of muscle tissue was homogenized in ice-cold lysis buffer (Cell Signaling, Beverly, MA, USA) using a commercially available micro-dismembrator (Braun, Melsungen, Germany). Homogenates were rotated for 60 min at 4 °C and centrifuged at 10,000g for 10 min at 4 °C, before obtaining the supernatants which represent the sarcoplasmic fractions. The protein content of each sample was determined by the Lowry test kit (BioRad Laboratories GmbH, Munich, Germany). Homogenates of each subject were diluted to a protein concentration determining 1.5 µg/µl homogenate. Homogenate of 25 µg protein was thawed on ice, suspended in buffer (0.5 M Tris-HCl, 10 % glycerol, 2 % sodium dodecyl sulphate, 5 % 2-mercaptoethanol and 0.05 % bromophenolblue) and heated at 95 °C for 7 min. Equal amounts of proteins (22.5 µg) for each subject and time point were separated on a 26-well, 4–12 % BIS-TRIS Gel using a gel casting system working with MOPS or MES electrophoresis buffer. Buffers and equipment were all from BIORAD (BioRad Laboratories GmbH, Munich, Germany). After electrophoretic separation, proteins were transferred to a polyvinylidene difluoride (PVDF) membrane (BioRad, Munich, Germany) by using a semidry blotting system (65 min at 12 V, 155 mA and 3 W, PerfectBlue Semidry Electro Blotter, Sedec S, PeqLab, Erlangen, Germany). Membranes were blocked in 5 % low-fat milk for 75 min. Subsequently, membranes were washed with Tris-buffered saline containing Tween (150 mM NaCl, 10 mM Tris-HCl, 0.05 % Tween-20, pH 7.5). Membranes were incubated overnight at 4 °C

with primary antibodies raised against phospho-specific and total proteins. The following antibodies were used: Cell Signaling Technologies (Cell Signaling, Beverly, MA, USA), total mTOR (1:1,500); p70s6k^{Thr398} (1:1,500), p70s6k^{Thr421/Ser424} (1:1,500), total p70s6k (1:1,000), pTSC2^{Thr1462} (1:2,000), pGSK3- β ^{Ser9} (1:1,500), p4E-BP1^{Tyr37/46} (1:2,000), total 4E-BP1 (1:1,500), pS6^{Ser235/236} (1:1,500); total S6 (1:1,000); pJNK^{Thr183/Tyr185} (1:1,500), pAKT^{Ser473} (1:1,000), pAKT^{Thr308}, total AKT (1:1,000). pmTOR^{Ser2448} (1:2,000) were used from ABCAM (Abcam, Cambridge, UK). Total JNK (1:1,500) and α -tubulin (1:4,000) antibodies were from Sigma (Sigma-Aldrich, Steinheim, Germany). pFAK^{Tyr397} (1:2,000) antibodies were purchased from Millipore (Billerica, MA, USA).

Secondary antibodies (HRP-linked goat anti-rabbit and goat anti-mouse; Cell Signaling, Beverly, MA, USA) were diluted 1:8,000 in 0.05 M TBS-T containing 5 % milk protein. Membranes were incubated for 1 h at room temperature under constant agitation. Proteins were detected by an enhanced chemo-luminescence assay (ECL Kit, Amersham-Life Science, Buckinghamshire, UK) exposed to X-ray film (Kodak X-OMAT Engineering, Eastman Kodak Co., Rochester, NY). Total levels of proteins were investigated on a separate gel with equal protein loading and served as internal loading controls. Phospho-specific signals were normalized to total protein levels and in case of pFAK^{Tyr397} and pGSK3- β ^{Ser9} to α -tubulin. Samples of at least one subject of each exercise group ran on the same gel under equal conditions, and samples of at least three distinct subjects were loaded on one single gel. All western blots were quantified using ImageJ® software (National Institutes of Health, USA) [18].

Statistics

All data are expressed as means $\pm$ standard error of means (S.E.M.). Multifactorial ANOVA for repeated measures with the factors 'group', 'time point' and 'fibre type' was applied to determine the differences in the phosphorylation pattern of sarcoplasmic staining in myofibres and in Western blotting over time. In the case of a significant difference between factors, Tukey's post hoc test was used to test for differences between groups, time points and fibre type. Statistical significance was assumed at $P < 0.05$.

Results

Serum creatine kinase

Serum CK levels increased significantly ($P < 0.01$) in HIT (210 ± 81 to 249 ± 67 U/L), STD (220 ± 96 to 264 ± 65 U/L) and ECC (144 ± 54 to 238 ± 93 U/L) at 24-h post-RE compared

to baseline (Fig. 1a). At 24-h post-exercise, the relative increase in CK levels was significantly higher ($\approx 50\%$) in ECC group than in STD and HIT ($P < 0.05$).

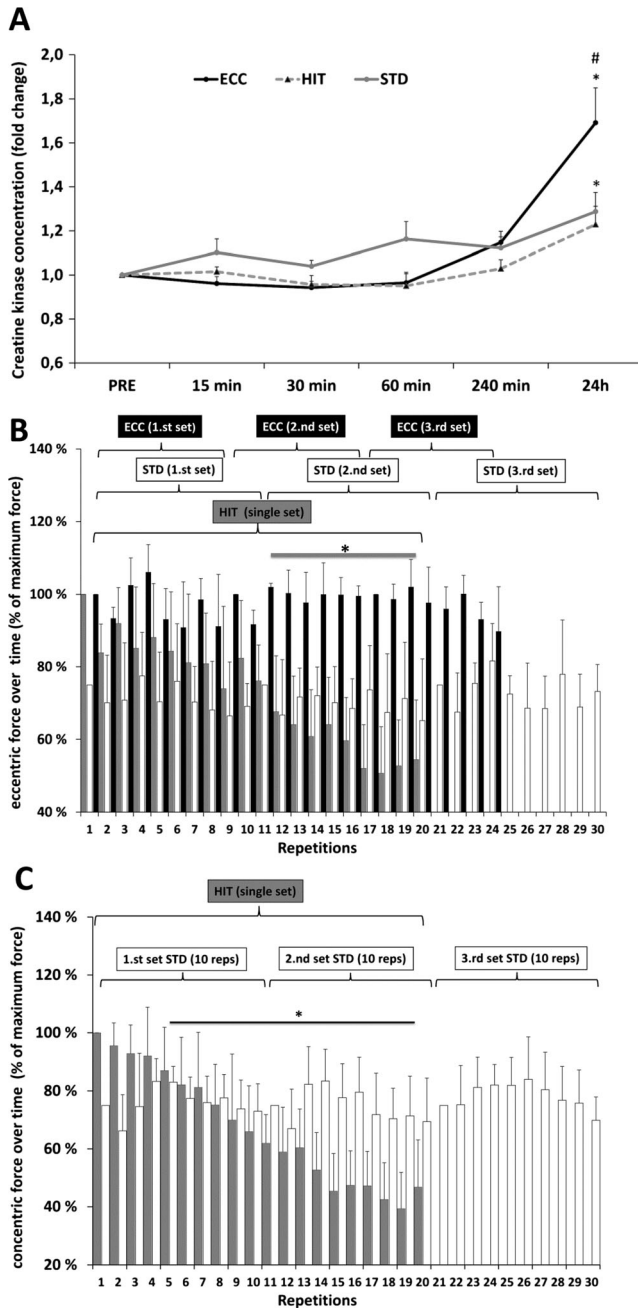


Fig. 1 **a** Serum creatine kinase levels (fold change from baseline values) in STD, HIT and ECC up to 24-h post-RE. * Significantly different compared with the baseline values in HIT, STD and ECC; # significantly different from STD and HIT in the ECC condition at that time point. **b** Percentage of eccentric force generation over time in relation to the first repetition in HIT, STD and ECC. * Significantly different from 1st repetition in HIT ($P < 0.05$, 11th and 12th repetition; $P < 0.01$, 14th to 20th repetition). **c** Percentage of concentric force generation over time in relation to the first repetition in HIT and STD. * Significantly different from 1st repetition in HIT ($P < 0.05$, 5th–7th repetition; $P < 0.01$, 8th–20th repetition)

Force generation

Neither eccentric (Fig. 1b) nor concentric force (Fig. 1c) dropped significantly in STD (75 % dynamic concentric and eccentric force) from the first to the last repetition during RE (concentric 190 ± 49 vs 175 ± 25 Nm, eccentric 214 ± 62 vs 226 ± 63 Nm) or ECC group (100 % dynamic eccentric force) from the first to the last repetition (eccentric 296 ± 69 vs 300 ± 71 Nm). In contrast, eccentric force dropped significantly in HIT (100 % of dynamic concentric and eccentric force) from the first repetition (287 ± 79 Nm) ($P < 0.05$) to the 11th repetition and below 55 % of initial force to the last repetition (176 ± 72 Nm) ($P < 0.01$). Concentric force decreased significantly in HIT from the 1st (279 ± 64 Nm) to the 6th repetition ($P < 0.05$) and was below 45 % of initial values to the last repetition (126 ± 44 Nm) ($P < 0.01$).

FAK

pFAK^{Tyr397} levels (Fig. 2a) were more than doubled at 30-min post-RE in ECC when compared to baseline conditions ($P < 0.01$) and significantly higher than those in STD and HIT to this time point ($P < 0.01$). However, also, HIT increased pFAK^{Tyr397} at 15 min after RE ($P < 0.05$) but with lower total activation levels. STD showed no increase in pFAK^{Tyr397} levels post-RE. Between 60-min and 24-h post-RE, pFAK^{Tyr397} levels decreased in all RE modes towards baseline levels.

JNK

pJNK54^{Thr183/Tyr185} levels (Fig. 2b) doubled at 15 min in STD and HIT ($P < 0.01$) and were still elevated 30-min post-RE ($P < 0.05$). Phosphorylation levels of JNK increased at 15 min 3.5-fold in ECC and were at this time point significantly higher than those in STD and HIT ($P < 0.05$). pJNK54^{Thr183/Tyr185} levels returned to near baseline values within 60-min post-RE in any group.

AKT

pAKT^{Thr308} levels (Fig. 2c) did not increase significantly in STD, increased 60-min post-RE in HIT ($P < 0.05$) and were 2-fold increased up to 240 min in ECC ($P < 0.05$). pAKT^{Thr308} increased significantly more at 15-min post-RE in ECC than HIT ($P < 0.01$) while no differences were detected between STD and ECC. pAKT^{Ser473} (Fig. 2d) increased in ECC up to 60-min post-RE ($P < 0.05$) but did not change in STD and HIT. pAKT^{Ser473} levels were significantly higher in ECC than those in HIT 15- and 30-min post-exercise ($P < 0.05$) but not different between ECC and STD to any time point. Increased AKT activity in ECC was supported by increased phosphorylation

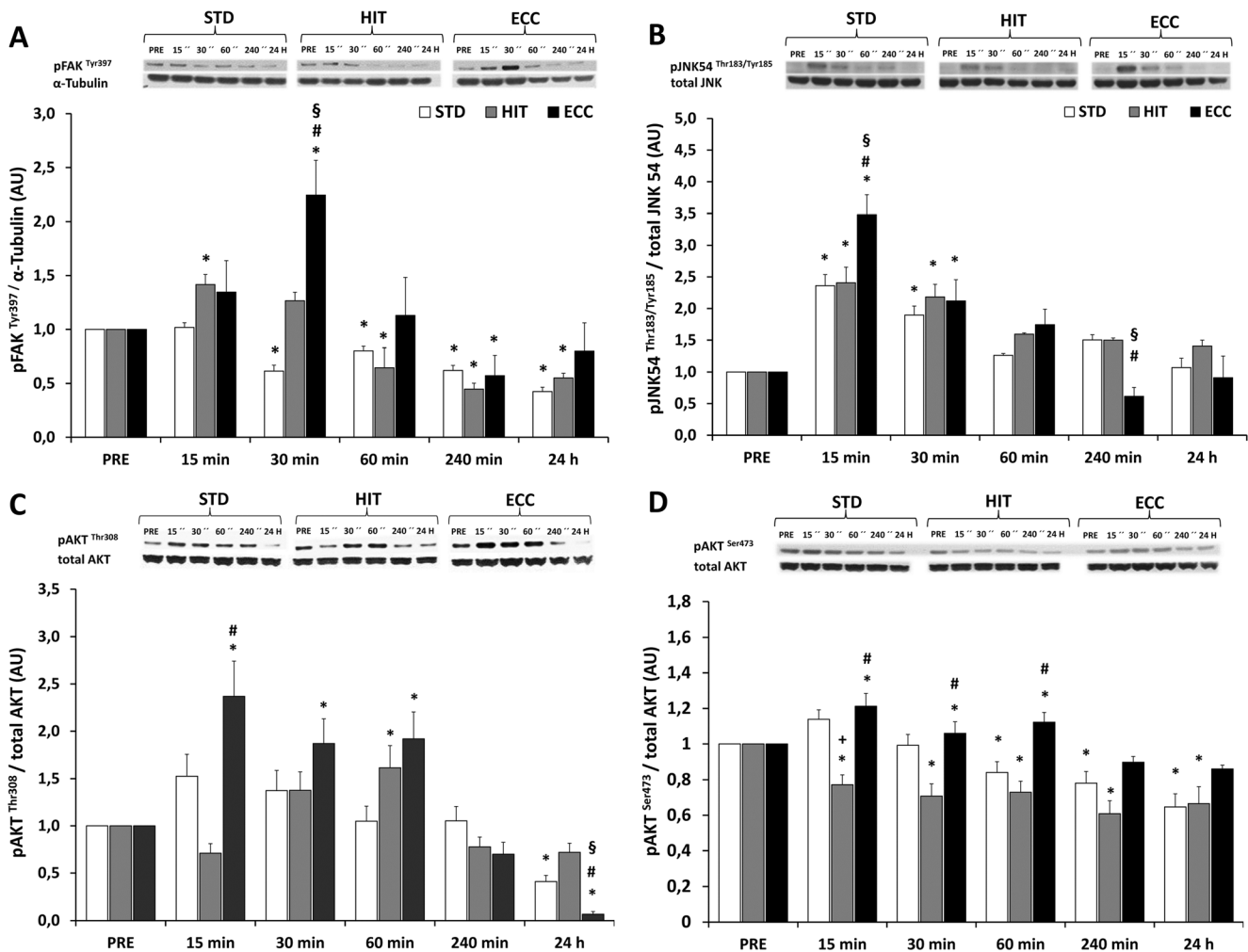


Fig. 2 **a** Ratio of phosphorylated pFAK^{Tyr397} to α -tubulin, **b** pJNK^{Thr183/Tyr185}, **c** pAKT^{Thr308} and **d** pAKT^{Ser473} to total levels in STD, HIT and ECC. Ratios are from rested fed baseline conditions and 15-, 30-, 60-, 240-min and 24-h after RE. Values are displayed as means $\pm$ S.E.M. Data are expressed in arbitrary units (AU). * Significantly different from PRE

within each exercise group, $P < 0.05$. $\S$ Significantly different from STD within that time point, $P < 0.05$. # Significantly different from HIT within that time point, $P < 0.05$. + Significantly different between HIT and STD within that time point, $P < 0.05$

of GSK3- β ^{Ser9} and TSC-2^{Thr1462} (see supplemental data Fig. 7).

mTOR

pmTOR^{Ser2448} levels (Fig. 3a) increased 2-fold in ECC at 30-min post-RE showing significantly increased levels compared to STD and HIT ($P < 0.05$, $P < 0.01$), respectively. HIT showed the weakest regulation of pmTOR^{Ser2448} compared to ECC and STD, and reduced levels compared to baseline were detected at 15-min post-RE ($P < 0.05$).

4E-BP1

p4E-BP1^{Thr37/46} levels (Fig. 3b) increased significantly in ECC (7-fold) compared to STD (2-fold) and HIT (1.8-fold)

up to 30-min post-RE ($P < 0.01$) and were also higher in STD than those in HIT 60 min following RE ($P < 0.01$).

p70s6k

Levels of p70s6k^{Thr421/Ser424} significantly increased up to 2.2-fold in every RE condition (Fig. 4a) post-RE. Peak signal intensity of p70s6k^{Thr421/Ser424} was equal in RE groups, and no differences over time between HIT, STD and ECC were detected.

p70s6k^{Thr389} levels significantly increased in all exercise modes over time (Fig. 4b) but were significantly higher at 60 min in ECC (3.5-fold) than STD (2.5-fold) ($P < 0.05$) and 15, 30, and 60 min upon stimulation in ECC than HIT (2-fold) ($P < 0.05$). p70s6k^{Thr389} levels displayed in HIT the weakest increase of all RE modes.

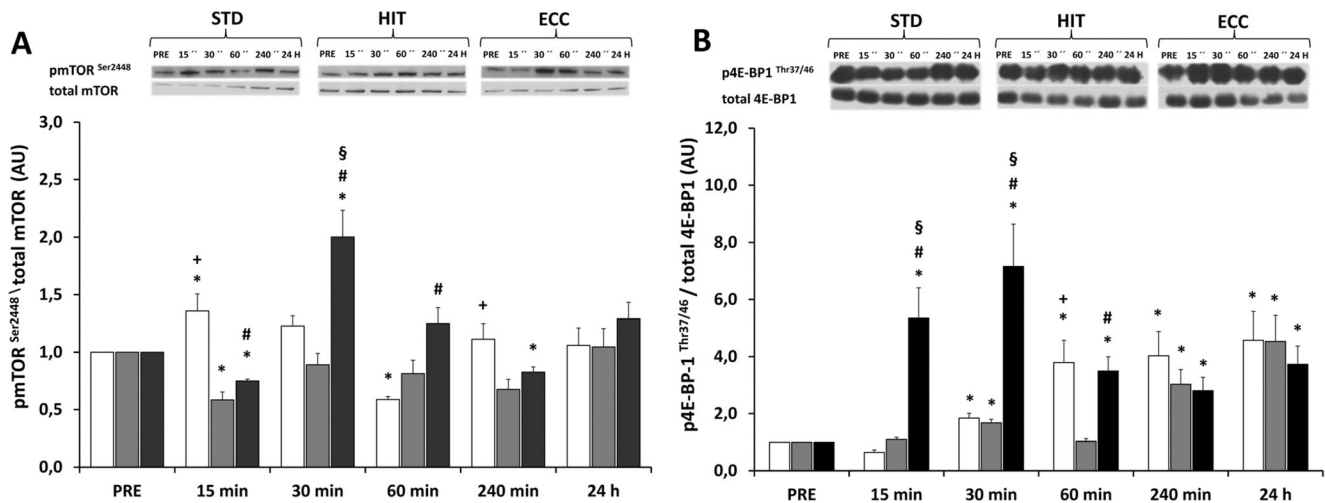


Fig. 3 **a** Ratio of phosphorylated pmTOR^{Ser2448} and **b** p4E-BP1^{Thr37/46} to total levels in STD, HIT and ECC. Ratios are from rested fed baseline conditions and 15-, 30-, 60-, 240-min and 24-h after RE. Values are displayed as means±S.E.M. Data are expressed in arbitrary units (AU).

* Significantly different from PRE within each exercise group, $P < 0.05$. § Significantly different from STD within that time point, $P < 0.05$. # Significantly different from HIT within that time point, $P < 0.05$. + Significantly different between HIT and STD within that time point, $P < 0.05$

S6

pS6^{Ser235/236} levels increased in all RE modes (Fig. 4c) and, however, showed a dominant increase in ECC over time. pS6 phosphorylation levels increased in ECC significantly higher (6-fold) than those in STD at 30 min ($P < 0.01$) and compared to HIT (2-fold) at 30- to 60-min post-RE (both $P < 0.01$). pS6^{Ser235/236} levels were significantly higher at 15-min post-RE in STD compared to HIT ($P < 0.01$).

Type I and II myofibre signalling

In order to localize fibre type specific differences in signalling responses, we determined phosphorylation of pJNK^{Thr183/Tyr185} (Fig. 5a, b) and pS6^{Ser235/236} (Fig. 5c–e) selectively in sarcoplasmic compartments of types I and II fibres of each RE group. pJNK^{Thr183/Tyr185} levels increased significantly in types I and II myofibres in each exercise group post-RE (Fig. 6a, c). In ECC, pJNK^{Thr183/Tyr185} levels increased in type I fibres significantly higher than those in type I fibres in HIT at 240-min post-RE ($P < 0.05$) and compared to STD at 60-min post-RE ($P < 0.01$). In type II fibres, pJNK^{Thr183/Tyr185} levels increased significantly more in ECC than those in HIT at 30-, 60- and 240-min post-RE ($P < 0.01$, $P < 0.05$) as well as 60 and 240 min in STD ($P < 0.01$). pJNK^{Thr183/Tyr185} levels were not different between types I and II myofibres in STD and HIT at any time point. However, higher levels of pJNK^{Thr183/Tyr185} were detected in type II than type I myofibres 30-min post-RE in ECC ($P < 0.05$).

We further determined increasing pS6^{Ser235/236} levels in each fibre type and training mode (Fig. 5c, d and Fig. 6b, d). pS6^{Ser235/236} staining in skeletal muscle myofibres showed in several cases clear-cut differences between pS6^{Ser235/236}

positive and negative fibres and a dominant staining of either type I or type II myofibres. Figure 5c pictures 3–4 display pS6^{Ser235/236}-stained myofibres of one subject from STD with a dominant staining in type I fibres. In contrast, Fig. 5d pictures 3–4 represent a dominant staining of type II myofibres of one subject from ECC.

Peak phosphorylation of S6 at Ser235/236 was elevated in type I myofibres up to 2.5-fold in ECC, 2.2-fold in STD and 2-fold in HIT. pS6^{Ser235/236} levels in type I fibres were higher in ECC than those in HIT at 30- and 60-min post-exercise ($P < 0.01$), but no differences between ECC and STD were detected. STD showed higher levels of pS6^{Ser235/236} at 15-min post-RE than HIT ($P < 0.01$). pS6^{Ser235/236} levels in type II fibres were significantly higher in ECC compared to STD at 30- and 240-min post-RE ($P < 0.05$, $P < 0.01$) and also compared to HIT from 30 to 240 min after RE ($P < 0.01$ – $P < 0.05$). Specifically in ECC, pS6^{Ser235/236} levels in type II fibres were significantly higher than those in type I fibres to each time point post-RE ($P < 0.01$ – $P < 0.05$). In ECC, type I fibres displayed significantly higher levels ($P < 0.01$) of pS6^{Ser235/236} than type I fibres in HIT at 15- to 60-min post-RE. Especially, type II myofibres displayed a higher increase in pS6^{Ser235/236} levels in ECC than type II fibres in STD and HIT. The smallest increase in pS6^{Ser235/236} levels was detected in both myofibre types in HIT after RE.

Discussion

We hypothesized that the highest force development with equal TUT represents a key determinant of the magnitude of RE-induced signalling responses. Thus, the present study

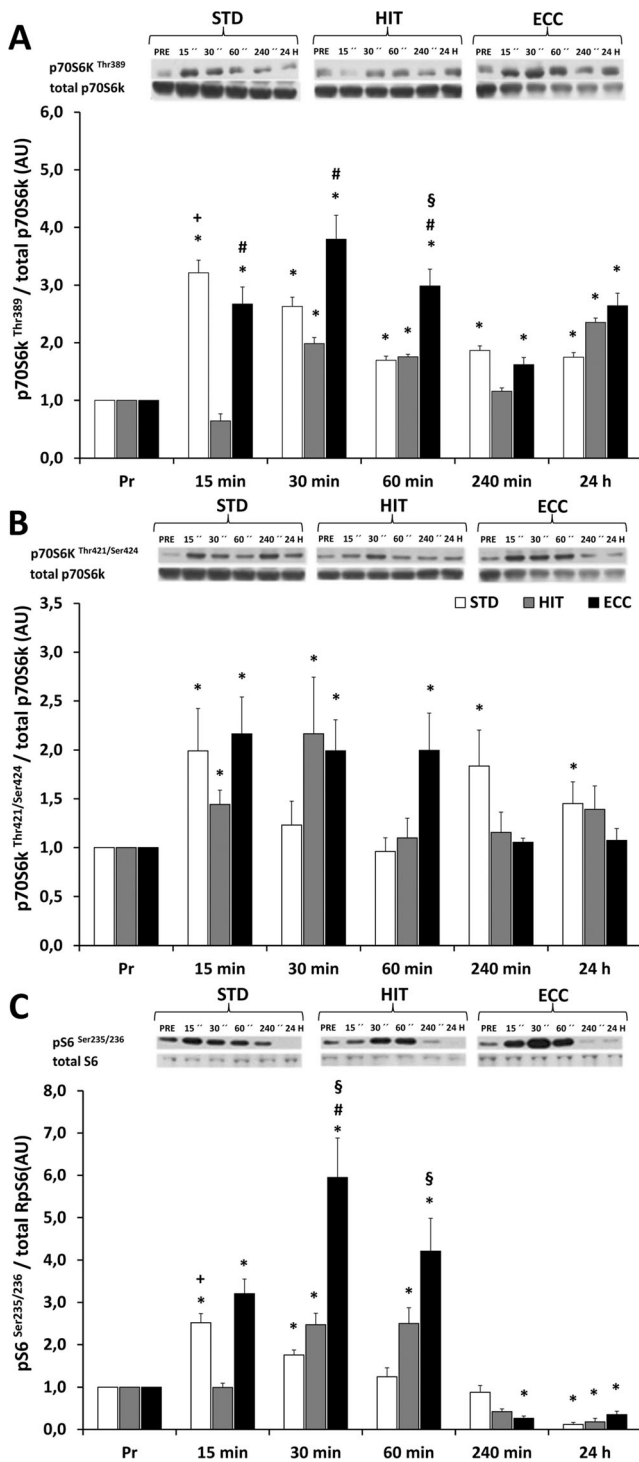


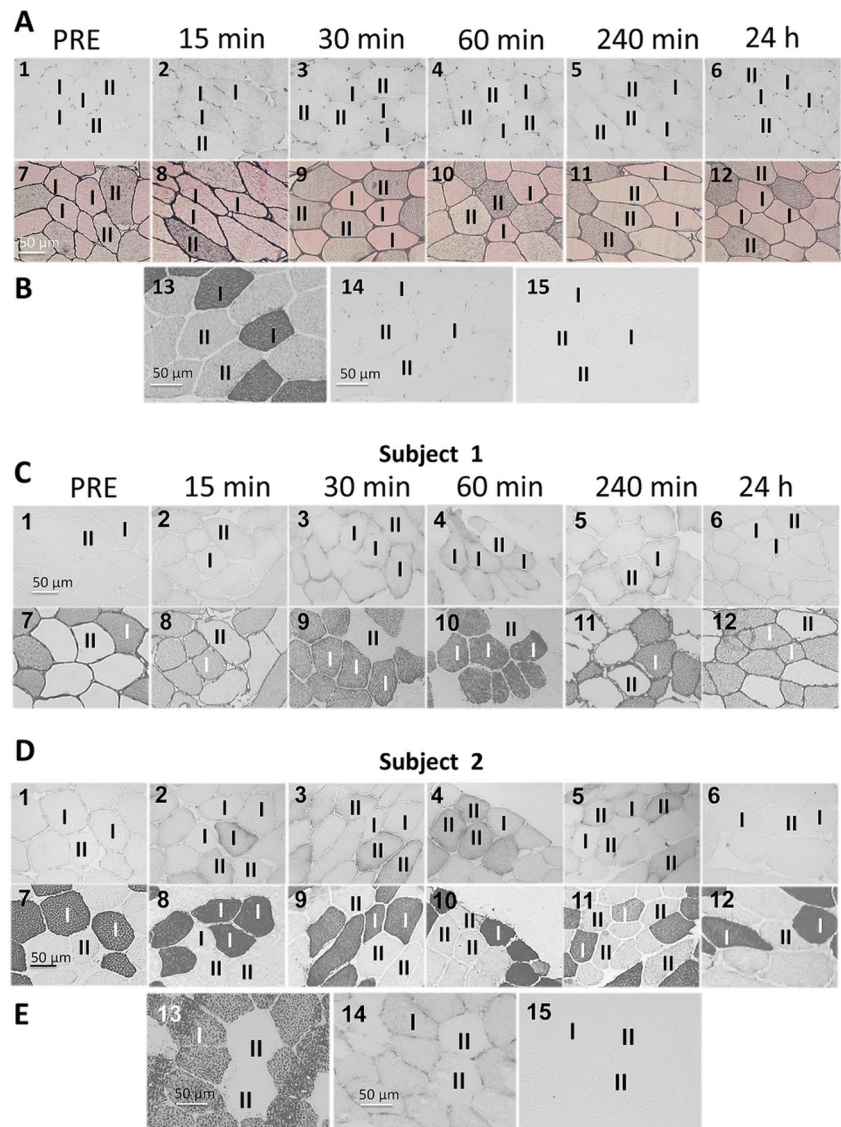
Fig. 4 **a** Ratio of phosphorylated p70s6k^{Ser421/Thr424}, **b** p70s6k^{Thr389} and **c** pS6^{Ser235/236} to total levels in STD, HIT and ECC. Ratios are from rested fed baseline conditions and 15-, 30-, 60-, 240- min and 24- h after RE. Values are displayed as means±S.E.M. Data are expressed in arbitrary units (AU). * Significantly different from PRE within each exercise group, $P<0.05$. # Significantly different from STD within that time point, $P<0.05$. § Significantly different from HIT within that time point, $P<0.05$. + Significantly different between HIT and STD within that time point, $P<0.05$

investigated acute molecular signalling in human skeletal muscle as initial and prolonged reaction to three modes of RE that were equalized in TUT. The highest signalling response was detected in subjects that performed ECC followed by STD, while the weakest response was observed in HIT subjects. In particular, because TUT was equalized in the applied exercise modes, and since the extent of the generated force over the time course differed between groups (Fig. 1b, c), this variable can be considered to be a major reason for the observed differences in signalling. While in STD and ECC, 3-min breaks between exercise bouts were applied, subjects in HIT were advised to perform 20 consecutive concentric and eccentric contractions with maximal effort. As a consequence, decreasing creatine phosphate levels or further factors that impair contractility of skeletal muscle [19] might have declined contractility in HIT over time, which led in sum to the lowest force generation in this exercise mode. By contrast, contractility in ECC was maintained, leading to the highest force development over time, significantly increased CK levels in ECC above STD and HIT at 24-h post-exercise, also indicating the highest mechanical strain in this exercise mode [20].

Mechanosensitive signalling

Increased mechanical demands of skeletal muscle induce integrin clustering and the integrin-dependent auto-phosphorylation of FAK at Tyr397, while MAPK kinases, particularly JNK, are increasingly phosphorylated in response to rising mechanical stress of myofibres [16]. Indeed, it has been shown that the time-tension integral is a key determinant of the magnitude of MAPK kinase activation and JNK phosphorylation [14]. Hence, in the present study, the significantly higher levels of pFAK^{Tyr397} and pJNK54^{Thr183/Tyr185} in ECC compared with those in HIT and STD reflect the highest mechanical demand on myofibres over TUT. Eccentric contractions require unique innervation patterns [21, 22] and the dynamic derecruitment of myofibres. This shifts skeletal muscle towards the enhanced recruitment of type II myofibres [17], explaining the significantly higher JNK phosphorylation levels in type II myofibres in ECC compared with those in STD and HIT. In cardiac and skeletal muscle, FAK is localized at costameres and hence at the very centre of force transmission at the sarcolemma [23]. FAK activation contributes through the phosphorylation of the mTOR downstream target p70s6k to increased protein synthesis, more importantly, by the circumvention of mTOR-related upstream signalling [15]. Hence, increased signalling via FAK might have played a role in the downstream responses of p70s6k, particularly in ECC in the present study. Although force development over time was a major difference between exercise modes and mechanosensitive markers offered the highest phosphorylation levels in ECC, also metabolic changes might have

Fig. 5 **a** Immunohistochemical staining of pJNK^{Thr183/Tyr185} in human skeletal muscle. *Pictures 1–6* show representative pictures ($\times 200$ magnification) of the localization and changing sarcoplasmic density of pJNK^{Thr183/Tyr185} over time. *Pictures 7–12* display consecutive cross sections stained for type I (black) and type II fibres (pale) for each time point ($\times 200$ magnification). **b** *Pictures 13–15* represent consecutive cross sections stained for type I myofibres (13), pJNK^{Thr183/Tyr185} (14) and an unstained negative control (15). **c, d** Immunohistochemical staining ($\times 200$ magnification) of pS6^{Ser235/236} stained myofibres of two different subjects. **c** *Pictures 1–6* depict myofibres of one representative subject from STD with predominantly type I myofibres displaying increased pS6^{Ser235/236} staining. **d** *Pictures 1–6* display one subject from ECC where predominantly type II myofibres show increased pS6^{Ser235/236} staining. **c, d** *Pictures 7–12* display consecutive cross sections for each time point stained for type I (black) and type II fibres (pale) ($\times 200$ magnification). **e** *Pictures 13–15* display consecutive cross sections stained for type I myofibres (13), pS6^{Ser235/236} (14) and the negative control (15)



contributed to distinct anabolic signalling. Both mechanical and metabolic perturbations are exerted simultaneously during contractions and converge within the signalling networks of skeletal muscle [24].

Signalling upstream of mTOR

We further determined the phosphorylation of several activity-related components of the mTOR pathway. A major upstream activation loop of the mTOR pathway depends on AKT activation by its phosphorylation at Thr308 via PDK1 [25] and via feed-forward activation at Ser473 by the Torc-2 complex of mTOR [26]. Both phosphorylation sites were significantly increased in ECC with only minor responses in STD and HIT. We conclude AKT to be sufficiently activated, particularly in ECC, since pGSK-3^{Ser9} and pTSC2^{Thr1462}, both known targets of AKT, showed increased phosphorylation in

ECC but only minor responses in STD and reduced phosphorylation in HIT (see [supplemental data](#)). Although the constitutive activation of AKT augments skeletal muscle hypertrophy [27], acute exercise induces partly increased [6, 8] or unchanged phosphorylation levels [9, 11, 28], and hence, AKT alone is a weak predictor of acute contractile stress. One reason for that might be the sensitivity of AKT towards various stimuli, especially nutrition [29, 30]. It has been demonstrated that increased insulin levels through glucose administration, but not RE, induces AKT phosphorylation, mTOR-related downstream signalling and elevated protein synthesis [29]. Because we disposed a small and standardized meal 2 h prior to RE and did not observe increased insulin levels in any group (data not shown), nutrition-based disparities might not have contributed to the differences in AKT phosphorylation between groups. Nonetheless, AKT has also been shown to be mechanosensitive [31], particularly because

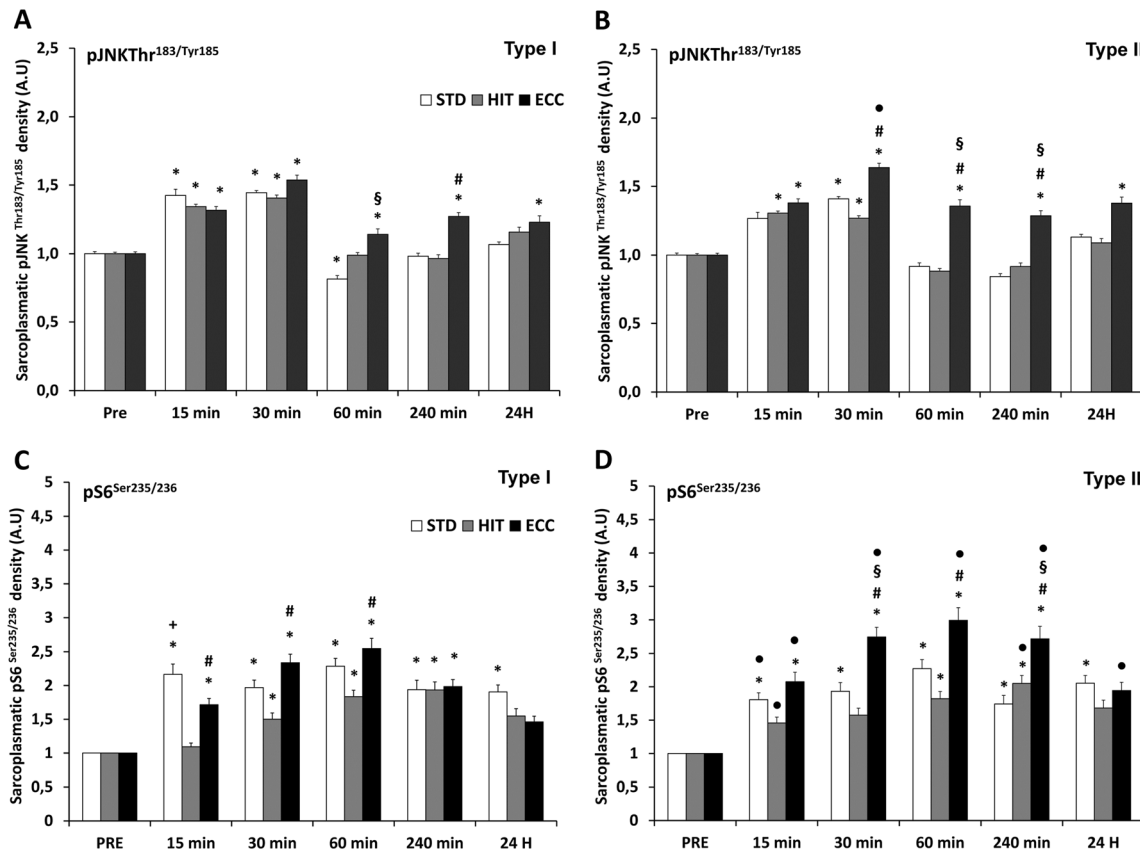


Fig. 6 **a, b** Quantitative analysis of the immunohistochemical staining of pJNK^{Thr183/Tyr185} and **c, d** pS6^{Ser235/236} in the sarcoplasmic compartments of type I and type II myofibers in STD, HIT and ECC. Ratios are from rested fed baseline conditions at 15-, 30-, 60-, 240-min and 24-h post-exercise. Values are displayed as means±S.E.M. Data are expressed in arbitrary units (AU). * Significantly different from PRE within each

exercise group and fibre type, $P < 0.05$. § Significantly different from STD within that time point and fibre type, $P < 0.05$. # Significantly different from HIT within that time point and fibre type, $P < 0.05$. + Significantly different between HIT and STD within that time point, $P < 0.05$. • Significantly different between type I and II myofibers within exercise groups to that time point, $P < 0.05$

phosphorylation at Ser473 increases in response to mechanical load in cardiac muscle [32], while AKT is also a potential target of FAK [33]. Another activator of AKT kinase is IGF-1, which has been demonstrated to be released via autocrine mechanisms in skeletal muscle cells [34] and via stretch-dependent mechanisms by the extracellular matrix [35]. Hence, all the mentioned mechanisms contribute to AKT activation, assuming that AKT is activated, especially by contractile load in the present study, which concurrently displays the major difference between our groups.

mTOR, as the central governor of protein synthesis and downstream element of AKT, was increasingly phosphorylated at Ser2448 in ECC, while in STD and HIT, phosphorylation levels were determined to be only weak or attenuated. However, mTOR phosphorylation can be regulated by various stimuli and independent of AKT [36]. Notably, the mechanical stimulation of skeletal muscle but also increased amino acid availability [37, 38] can increase mTOR activity. Consequently, mTOR phosphorylation in response to different exercise regimens, nutritional conditions and time courses of investigation post-exercise offers inconsistent responses with

increased [6, 39, 40] or unchanged phosphorylation [11, 41] and regardless of the exercise mode. Hence, the determination of RE-induced signalling must involve a row of upstream and downstream targets of mTOR-related components to depict differences in activity levels of signalling proteins occurring in dependence of the RE mode.

Signalling downstream of mTOR

4E-BP1 is directly phosphorylated by mTOR, which leads to its inhibition and release from the eIF4E complex, allowing translation initiation [42]. Since 4E-BP1 phosphorylation was also significantly higher in ECC than that in STD and HIT, this was likely to have been induced by increased mTOR activation in ECC. A further downstream target of mTOR is p70s6k, and its phosphorylation correlates with acutely increased protein synthesis and the hypertrophy of skeletal muscle [5, 43]. mTOR regulates the phosphorylation of p70s6k at Thr389, which was increased in all exercise modes in the present study. However, ECC offered the strongest response 30- and 60-min post-exercise, the time point at

which mTOR also showed the highest phosphorylation levels in ECC. However, only the coincident phosphorylation of p70s6k by mTOR at Thr389 and additionally at Thr421/Ser424 activates p70s6k. The regulation of p70s6k activity is controlled by a series of phosphorylation steps with Thr421/Ser424 in the auto-inhibitory domain preceding the further phosphorylation at Thr389 in the linker domain that activates the kinase [44]. p70s6k phosphorylation at Thr421/Ser424 occurred after all RE modes and at a similar magnitude, suggesting exercise-induced stimulation to be sufficient to mediate this response independent of exercise mode. p70s6k further critically regulates the phosphorylation of ribosomal protein S6 at Ser235/236, which is a key initiator of translation initiation [45]. As the phosphorylation of p70s6k occurred significantly at both sites in ECC, p70s6k activity might have been increased to the highest extent in this group. This finding may also explain why ribosomal protein S6 phosphorylation at Ser235/236 increased in ECC significantly higher than it did in STD and HIT. However, surprisingly, we also detected the increased phosphorylation of S6 in HIT despite the blunted signalling of AKT and mTOR. This observation was unexpected when adopting the signal transduction via mTOR to be vital for the induction of S6 phosphorylation in loaded skeletal muscle. However, the phosphorylation of p70s6k was also observed in HIT at both residues without increased mTOR phosphorylation. Hence, the activation of p70s6k was sufficiently activated in this group independent of mTOR and mTOR-related upstream signalling. This finding might be explained by the FAK-dependent activation of p70s6k, which we also observed in HIT at 15-min post-exercise and which might have been induced by the initial intense repetitions in HIT despite the significant reductions in force until the end of the exercise. Additionally, MAPK kinase signalling via ERK and p90s6k (RSK1) contributes to S6 phosphorylation [46]. Although ERK phosphorylation was not determined in the present study, this might be speculated since ERK activation is sensitive to mechanically induced stress [16, 47], and its activity has been shown to increase after RE [48].

Fibre type specific signalling

We determined specific and significant differences in pS6^{Ser235/236} staining between myofibre types and exercise modes. In particular, type II myofibres in ECC showed significantly higher pS6^{Ser235/236} levels than that in type I fibres in ECC as well as type II fibres in STD and HIT. This finding was accompanied by a similar response of JNK phosphorylation in myofibres. However, in contrast to pJNK^{Thr183/Tyr185}, pS6^{Ser235/236} levels showed clear-cut differences between weak and strongly stained myofibres (Fig. 5c, d; pictures 3–4) assuming that this difference was induced as a consequence of reduced or increased activity of motoneurons innervating these fibres during RE. It has to be considered that Western

blotting indicates augmented signalling, when more myofibres in the lysate display increased phosphorylation of a signalling protein. Hence, it can be assumed that specifically in ECC, increased pJNK54^{Thr183/Tyr185} and pS6^{Ser235/236} levels in single or in more myofibres, respectively, contributed to the observed increases in Western blotting signals due to enhanced recruitment and mechanical strain of single myofibres compared with STD and HIT. In this regard, Goodman and colleagues have illustrated with the novel and elegant ‘SUnSET’ approach [49] via immunohistochemistry [50] that differences in fibre type specific protein synthesis can precisely be detected and that these distinctions contribute to variations in whole skeletal muscle responses of muscle growth.

General significance

The present study showed that ECC induces the highest phosphorylation of signalling proteins that regulate skeletal muscle adaptation. Importantly, this dominant response occurred with equal TUT compared with STD and HIT. Because the phosphorylation of p70s6k^{Ser421/424} and p70s6k^{Thr389}, p4E-BP1^{Thr37/46} and pS6^{Ser235/236} directly contributes to regulate protein synthesis, it can be speculated that ECC exercise might have induced the highest potential for protein synthetic responses and hence muscle growth in the present study. Owing to technical limitations, we were not able to measure protein synthesis directly or indirectly which is why we have to speculate about this event. However, this assumption might be strengthened by studies in single fibres [49] or whole skeletal muscle [51] where anabolic signalling of pS6^{Ser235/236} has been shown to be concomitantly increased with protein synthesis under growth promoting conditions, while down-regulation of both was observed under conditions of muscle atrophy [51]. To date, several studies have found good relations between anabolic signalling, protein synthesis and hypertrophy [5, 7, 13], while others discuss this relation [39, 52, 53]. This can be caused by several reasons. For example, the phosphorylation of signalling proteins can be reduced in response to long-term training what might weaken this relation [52]. However, it can be assumed that also protein synthesis will be reduced under these conditions. More important, in a broad number of different studies, timing of biopsies, nutrition, subjects and exercise modes show considerable differences, why relations between acute signalling and protein synthesis can be blurred. In the present study, e.g. pS6^{Ser235/236} levels were reduced by 2–3-fold from 30- to 60-min post-exercise what would modulate the correlation between synthesis and signalling. Thus, it is important to consider timing of measurements to depict clear relations between faster and slower components of signalling and protein synthesis.

Although the extrapolation of acute signalling associated with protein synthesis towards long-term hypertrophy has to be carried out carefully, the standardized comparison of

distinct RE modes via essential signalling measures allowed us to assess the acute efficiency of distinct force development and TUT in inducing early anabolic signalling responses which regulate protein synthesis in skeletal muscle. The present study adds important knowledge to the understanding of RE-induced skeletal muscle responses, as we demonstrate that high force generation over time is an important trigger of RE-induced signalling in particular with an emphasis on enhanced type II fibre responses. We particularly assume enhanced myofibre recruitment and increased mechanical stimulation of myofibres to have induced augmented signalling via mechanosensitive and growth-related signalling pathways which, in interplay, depict a clear and distinct signalling pattern of RE-induced skeletal muscle activation in response to different loading conditions. Besides arguments that may contradict the long-term advantage of solely eccentric exercise on muscle growth (e.g. higher levels of protein degradation), this RE mode may exert beneficial effects in emphasizing maximal skeletal muscle activation in a limited training time and for selected time points during a training intervention. Further, the directed activation of type II fibres via eccentric exercise or RE modes with high force development can exert beneficial effects under conditions of chronically reduced activation in the elderly and after immobilization. The present study further elucidates that in some conditions, high muscle fatigue, as a subjective proxy for maximal muscle adaptation, may indeed exert blunted post-exercise-induced signalling via reduced fibre recruitment and the loss of force generation capacity. This finding is of considerable importance for exercise conception, especially when compared to other RE modes with an equal TUT of skeletal muscle.

Acknowledgments The authors thank Katharina Hermanns and the technicians Anika Voss, Bianca Collins and Mojghan Ghilav (Institute of Cardiovascular Research and Sport Medicine, Department of Molecular and Cellular Sport Medicine) for expert technical assistance.

Conflict of interest All authors disclose any professional relationship with companies or manufacturers who will benefit from the results of the present study.

Funding This investigation was funded by the Federal Institute of Sports Science (BISP) IIA1-507 070103/09-10. The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

References

1. Miller BF, Olesen JL, Hansen M, Døssing S, Cramer RM, Welling RJ, Langberg H, Flyvbjerg A, Kjaer M, Babraj JA, Smith K, Rennie MJ (2005) Coordinated collagen and muscle protein synthesis in human patella tendon and quadriceps muscle after exercise. *J Physiol* 567:1021–1033
2. Drummond MJ, Fry CS, Glynn EL, Dreyer HC, Dhanani S, Timmerman KL, Volpi E, Rasmussen BB (2009) Rapamycin administration in humans blocks the contraction-induced increase in skeletal muscle protein synthesis. *J Physiol* 587:1535–1546
3. Inoki K, Ouyang H, Zhu TQ, Lindvall C, Wang Y, Zhang XJ, Yang Q, Bennett C, Harada Y, Stankunas K, Wang CY, He X, MacDougall OA, You M, Williams BO, Guan KL (2006) TSC2 integrates Wnt and energy signals via a coordinated phosphorylation by AMPK and GSK3 to regulate cell growth. *Cell* 126:955–968
4. Kubica N, Bolster DR, Farrell PA, Kimball SR, Jefferson LS (2005) Resistance exercise increases muscle protein synthesis and translation of eukaryotic initiation factor 2B epsilon mRNA in a mammalian target of rapamycin-dependent manner. *J Biol Chem* 280:7570–7580
5. Baar K, Esser K (1999) Phosphorylation of p70(S6k) correlates with increased skeletal muscle mass following resistance exercise. *Am J Physiol* 276:C120–127
6. Dreyer HC, Fujita S, Cadenas JG, Chinkes DL, Volpi E, Rasmussen BB (2006) Resistance exercise increases AMPK activity and reduces 4E-BP1 phosphorylation and protein synthesis in human skeletal muscle. *J Physiol Lond* 576:613–624
7. Terzis G, Georgiadis G, Stratakis G, Vogiatzis I, Kavouras S, Manta P, Mascher H, Blomstrand E (2008) Resistance exercise-induced increase in muscle mass correlates with p70S6 kinase phosphorylation in human subjects. *Eur J Appl Physiol* 102:145–152
8. Camera DM, Edge J, Short MJ, Hawley JA, Coffey VG (2010) Early time course of Akt phosphorylation after endurance and resistance exercise. *Med Sci Sports Exerc* 42:1843–1852
9. Tannerstedt J, Apro W, Blomstrand E (2009) Maximal lengthening contractions induce different signaling responses in the type I and type II fibers of human skeletal muscle. *J Appl Physiol* 106:1412–1418
10. Bird SP, Tarpenning KM, Marino FE (2005) Designing resistance training programmes to enhance muscular fitness: a review of the acute programme variables. *Sports Med* 35:841–851
11. Eliasson J, Elfegoun T, Nilsson J, Köhne R, Ekblom B, Blomstrand E (2006) Maximal lengthening contractions increase p70 S6 kinase phosphorylation in human skeletal muscle in the absence of nutritional supply. *Am J Physiol Endocrinol Metab* 291:E1197–1205
12. Terzis G, Spengos K, Mascher H, Georgiadis G, Manta P, Blomstrand E (2010) The degree of p70(S6k) and S6 phosphorylation in human skeletal muscle in response to resistance exercise depends on the training volume. *Eur J Appl Physiol* 110:835–843
13. Burd NA, West DW, Staples AW, Atherton PJ, Baker JM, Moore DR, Holwerda AM, Parise G, Rennie MJ, Baker SK, Phillips SM (2010) Low-load high volume resistance exercise stimulates muscle protein synthesis more than high-load low volume resistance exercise in young men. *PLoS One* 5:e12033
14. Martineau LC, Gardiner PF (2002) Skeletal muscle is sensitive to the tension-time integral but not to the rate of change of tension, as assessed by mechanically induced signaling. *J Biomech* 35:657–663
15. Klossner S, Durieux AC, Freyssen D, Flueck M (2009) Mechanotransduction to muscle protein synthesis is modulated by FAK. *Eur J Appl Physiol* 106:389–398
16. Martineau LC, Gardiner PF (2001) Insight into skeletal muscle mechanotransduction: MAPK activation is quantitatively related to tension. *J Appl Physiol* 91:693–702
17. Nardone A, Schieppati M (1988) Shift of activity from slow to fast muscle during voluntary lengthening contractions of the triceps surae muscles in humans. *J Physiol* 395:363–381
18. Gehlert S, Bungartz G, Willkomm L, Korkmaz Y, Pfannkuche K, Schiffer T, Bloch W, Suhr F (2012) Intense resistance exercise induces early and transient increases in ryanodine receptor 1 phosphorylation in human skeletal muscle. *PLoS One* 7:e49326
19. Allen DG, Lamb GD, Westerblad H (2008) Skeletal muscle fatigue: cellular mechanisms. *Physiol Rev* 88:287–332
20. Fridén J, Sjöström M, Ekblom B (1983) Myofibrillar damage following intense eccentric exercise in man. *Int J Sports Med* 4:170–176

21. Enoka RM (1996) Eccentric contractions require unique activation strategies by the nervous system. *J Appl Physiol* 81:2339–2346
22. Nardone A, Romanò C, Schieppati M (1989) Selective recruitment of high-threshold human motor units during voluntary isotonic lengthening of active muscles. *J Physiol* 409:451–471
23. Flück M, Ziemiecki A, Billetter R, Müntener M (2002) Fibre-type specific concentration of focal adhesion kinase at the sarcolemma: influence of fibre innervation and regeneration. *J Exp Biol* 205:2337–2348
24. Matsakas A, Patel K (2009) Intracellular signalling pathways regulating the adaptation of skeletal muscle to exercise and nutritional changes. *Histol Histopathol* 24:209–222
25. Glass DJ (2003) Signalling pathways that mediate skeletal muscle hypertrophy and atrophy. *Nat Cell Biol* 5:87–90
26. Sarbassov DD, Guertin DA, Ali SM, Sabatini DM (2005) Phosphorylation and regulation of Akt/PKB by the rictor-mTOR complex. *Science* 307:1098–1101
27. Blaauw B, Canato M, Agatea L, Toniolo L, Mammucari C, Masiero E, Abraham R, Sandri M, Schiaffino S, Reggiani C (2009) Inducible activation of Akt increases skeletal muscle mass and force without satellite cell activation. *Faseb Journal* 23:3896–3905
28. Koopman R, Zorenc AHG, Gransier RJJ, Cameron-Smith D, van Loon LJC (2006) Increase in S6K1 phosphorylation in human skeletal muscle following resistance exercise occurs mainly in type II muscle fibers. *Am J Physiol Endocrinol Metab* 290:E1245–E1252
29. Greenhaff PL, Karagounis LG, Peirce N, Simpson EJ, Hazell M, Layfield R, Wackerhage H, Smith K, Atherton P, Selby A, Rennie MJ (2008) Disassociation between the effects of amino acids and insulin on signaling, ubiquitin ligases, and protein turnover in human muscle. *Am J Physiol Endocrinol Metab* 295:E595–604
30. Liao Y, Hung M-C (2010) Physiological regulation of Akt activity and stability. *Am J Transl Res* 2:19–42
31. Blaauw B, Mammucari C, Toniolo L, Agatea L, Abraham R, Sandri M, Reggiani C, Schiaffino S (2008) Akt activation prevents the force drop induced by eccentric contractions in dystrophin-deficient skeletal muscle. *Hum Mol Genet* 17:3686–3696
32. Kim CH, Cho YS, Chun YS, Park JW, Kim MS (2002) Early expression of myocardial HIF-1 α in response to mechanical stresses: regulation by stretch-activated channels and the phosphatidylinositol 3-kinase signaling pathway. *Circ Res* 90:E25–33
33. Franchini KG, Torsoni AS, Soares PH, Saad MJ (2000) Early activation of the multicomponent signaling complex associated with focal adhesion kinase induced by pressure overload in the rat heart. *Circ Res* 87:558–565
34. Jones JI, Gockerman A, Busby WH, Camacho-Hubner C, Clemmons DR (1993) Extracellular matrix contains insulin-like growth factor binding protein-5: potentiation of the effects of IGF-I. *J Cell Biol* 121:679–687
35. Perrone CE, Fenwick-Smith D, Vandenberg HH (1995) Collagen and stretch modulate autocrine secretion of insulin-like growth factor-I and insulin-like growth factor binding proteins from differentiated skeletal muscle cells. *J Biol Chem* 270:2099–2106
36. Philp A, Hamilton DL, Baar K (2011) Signals mediating skeletal muscle remodeling by resistance exercise: PI3-kinase independent activation of mTORC1. *J Appl Physiol* 110:561–568
37. Hornberger TA, Chu WK, Mak YW, Hsiung JW, Huang SA, Chien S (2006) The role of phospholipase D and phosphatidic acid in the mechanical activation of mTOR signaling in skeletal muscle. *Proc Natl Acad Sci U S A* 103:4741–4746
38. You JS, Frey JW, Hornberger TA (2012) Mechanical stimulation induces mTOR signaling via an ERK-independent mechanism: implications for a direct activation of mTOR by phosphatidic acid. *PLoS One* 7:e47258
39. Mitchell CJ, Churchward-Venne TA, West DW, Burd NA, Breen L (1985) Baker SK, and Phillips SM (2012) Resistance exercise load does not determine training-mediated hypertrophic gains in young men. *J Appl Physiol* 113:71–77
40. Moore DR, Atherton PJ, Rennie MJ, Tarnopolsky MA, Phillips SM (2011) Resistance exercise enhances mTOR and MAPK signalling in human muscle over that seen at rest after bolus protein ingestion. *Acta Physiologica* 201:365–372
41. Glover EI, Oates BR, Tang JE, Moore DR, Tarnopolsky MA, Phillips SM (2008) Resistance exercise decreases eIF2B ϵ phosphorylation and potentiates the feeding-induced stimulation of p70S6K1 and rpS6 in young men. *Am J Physiol Regul Integr Comp Physiol* 295:R604–610
42. Shah OJ, Anthony JC, Kimball SR, Jefferson LS (2000) 4E-BP1 and S6K1: translational integration sites for nutritional and hormonal information in muscle. *Am J Physiol Endocrinol Metab* 279:E715–E729
43. Phillips SM, Tipton KD, Aarsland A, Wolf SE, Wolfe RR (1997) Mixed muscle protein synthesis and breakdown after resistance exercise in humans. *Am J Physiol Endocrinol Metab* 273:E99–E107
44. Pullen N, Thomas G (1997) The modular phosphorylation and activation of p70s6k. *FEBS Lett* 410:78–82
45. Ruvinsky I, Meyuhas O (2006) Ribosomal protein S6 phosphorylation: from protein synthesis to cell size. *Trends Biochem Sci* 31:342–348
46. Roux PP, Shahbazian D, Vu H, Holz MK, Cohen MS, Taunton J, Sonenberg N, Blenis J (2007) RAS/ERK signaling promotes site-specific ribosomal protein S6 phosphorylation via RSK and stimulates cap-dependent translation. *J Biol Chem* 282:14056–14064
47. Csukly KJ, Martineau LC, Gardiner PF (2002) Inter- and intra-muscle comparisons of MAPK mechanosensitivity: evidence for the absence of fibre-type dependency. *Pflugers Arch* 444:732–737
48. Williamson D, Gallagher P, Harber M, Hollon C, Trappe S (2003) Mitogen-activated protein kinase (MAPK) pathway activation: effects of age and acute exercise on human skeletal muscle. *Journal of Physiology-London* 547:977–987
49. Goodman CA, Kotecki JA, Jacobs BL, Hornberger TA (2012) Muscle fiber type-dependent differences in the regulation of protein synthesis. *PLoS One* 7:e37890
50. Goodman CA, Mabrey DM, Frey JW, Miu MH, Schmidt EK, Pierre P, Hornberger TA (2011) Novel insights into the regulation of skeletal muscle protein synthesis as revealed by a new nonradioactive in vivo technique. *FASEB J* 25:1028–1039
51. Bentzinger CF, Lin S, Romanino K, Castets P, Guridi M, Summermatter S, Handschin C, Tintignac LA, Hall MN, Rüegg MA (2013) Differential response of skeletal muscles to mTORC1 signaling during atrophy and hypertrophy. *Skelet Muscle* 3:6
52. Mayhew DL, Kim JS, Cross JM, Ferrando AA, Bammann MM (2009) Translational signaling responses preceding resistance training-mediated myofiber hypertrophy in young and old humans. *J Appl Physiol* 107:1655–1662
53. West DWD, Kujbida GW, Moore DR, Atherton P, Burd NA, Padzik JP, De Lisio M, Tang JE, Parise G, Rennie MJ, Baker SK, Phillips SM (2009) Resistance exercise-induced increases in putative anabolic hormones do not enhance muscle protein synthesis or intracellular signalling in young men. *Journal of Physiology-London* 587:5239–5247